

AI BLAST SAFETY ASSISTANT

VERSION 1.0
OPERATIONS MANUAL

Official User Operating & System Administration Manual

[CHAPTER 1] System Access & Secure Authentication

To launch the system, access **http://localhost:5173/** in Google Chrome or Microsoft Edge. Sign in using the prefilled **Blasting Officer credentials** (Username: **officer** / Password: **officer123**). The top navigation bar will confirm your active session level (**OFFICER Session**).

[CHAPTER 2] Module 1 — Pre-Blast Safety Checklist Workflow

- Input Parameters:** Fill in site details (Site Name, Blast ID). The **Blast Date** automatically defaults to the present date.
- Check 17 Parameters:** Verify Weather (Temperature, Rain, Wind, Lightning), Workforce, Equipment, and Site parameters.
- Evaluate Risk:** Click **EVALUATE SAFETY RISKS**. The deterministic rule engine computes a score (0-100) and displays a **GREEN / YELLOW / ORANGE / RED** beacon badge.
- AI Safety Advisory:** View recommendations generated by **Google Gemini 3.1 Flash-Lite** (indicated by the green **AI PREDICTED (GEMINI 3.1)** badge).

[CHAPTER 3] Blasting Officer Sign-Off & Single-Page PDF Audit Generation

- Sign-Off Authorization:** Blasting Officers capture their digital signature on the screen canvas, enter comments, and select **APPROVE**, **HOLD**, or **REJECT**.
- Single-Page PDF Export:** Click **DOWNLOAD TAMPER-EVIDENT PDF REPORT**. The system builds a single-page A4 audit report featuring side-by-side verification tables, your signature graphic, and a **SHA-256 document verification hash**.

[CHAPTER 4] Module 2 — Blast Pattern Design & Fragmentation Simulation

Switch to the **Blast Design Optimisation** tab. Enter burden, spacing, bench height, and explosive mass. Click **GENERATE PLAN & RUN SIMULATION** to compute **Ground Vibration (PPV mm/s)**, **Air Blast Noise (dB)**, **Flyrock Risk**, and view 2D pattern layouts alongside the Rosin-Rammler fragmentation curve.

[CHAPTER 5] Module 3 — Incident Registers & Safety Auditing

Switch to the **Incident Registers & Audits** tab. Log active site infractions or equipment failures. All logged incidents are permanently recorded in the database and linked to pre-blast inspection reports.

[CHAPTER 6] System Administration & Environment Configuration

Environment settings are managed in **backend/.env**:

- API Credentials:** Set **ANTHROPIC_API_KEY** to your Google Gemini or Anthropic API key.
- Server Startup:** Run backend using **python main.py** (Port 8000) and frontend using **npm run dev** (Port 5173).